

Limb Proportions and Locomotion

Outline

Introduction

- Locomotor diversity and modes of locomotion in primates
- Review bones of interest

Activity (part 1)

- Locomotion videos: Make observations and form hypotheses
- Brainstorm: How can we calculate limb proportions to test our hypotheses?
- Finalize hypotheses

Mystery specimens and data collection protocol

- How to measure limb bone lengths

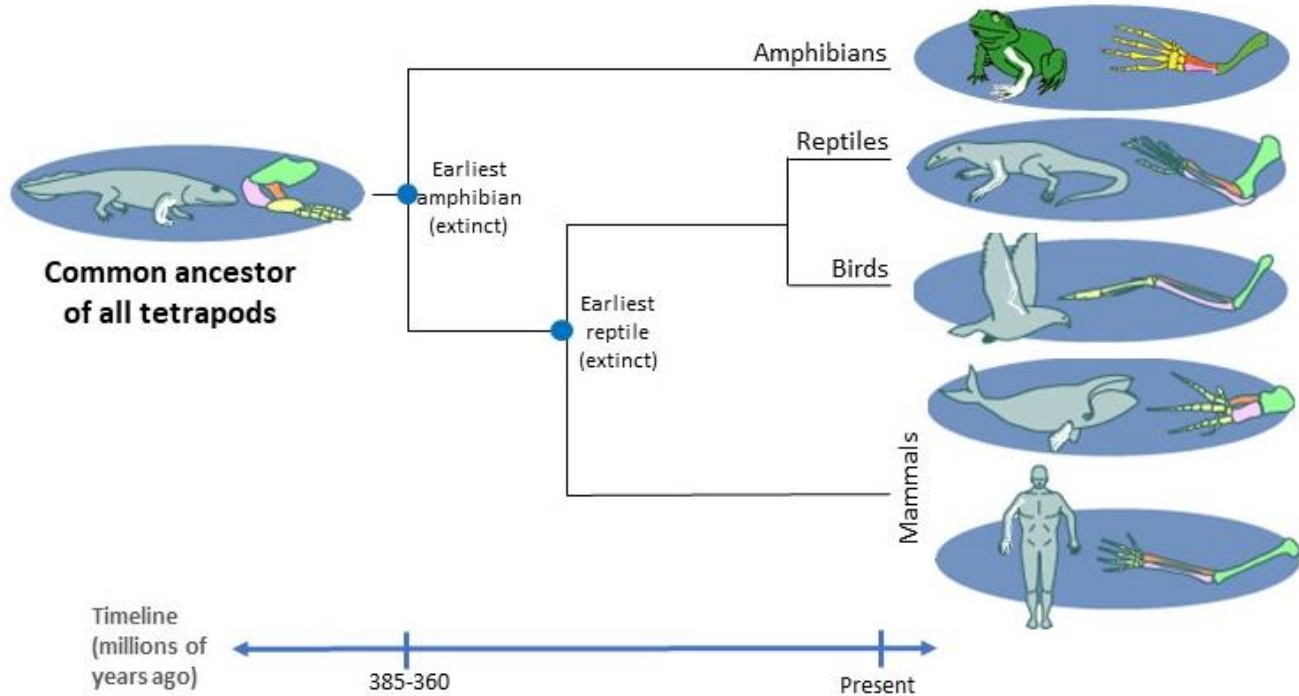
After this:

- Morphosource tutorial with Sharon
- Graphing and interpreting your data
- Graphing tutorial in Plotly

Evolutionary origins of 4-legged vertebrates

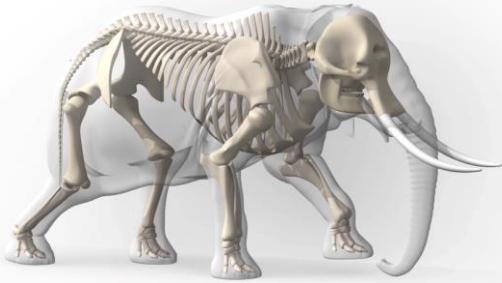
The limb structure of amphibians, reptiles, birds, and mammals was passed down from a 4-legged ancestor (**tetrapods**)

→ Same bones, different morphology



African Elephant

Mammal



Cheetah

Mammal



Tiger Salamander

Amphibian



Zebra-tailed Lizard

Reptile

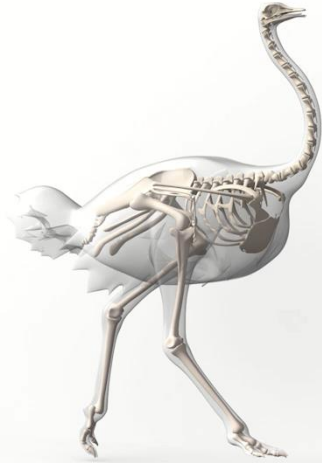


Quadrupedal
walking and
running

*Quadrupedal = locomoting
on all 4 legs*

Ostrich

Bird



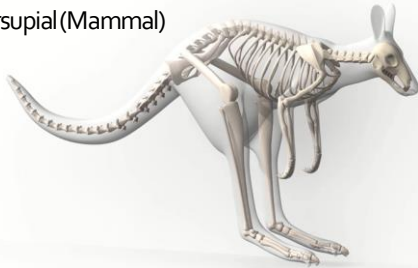
Human

Mammal



Kangaroo

Marsupial (Mammal)



Bipedal
walking and
hopping

*Bipedal = locomoting on 2
legs*

Gibbon

Mammal



Grey Squirrel

Mammal



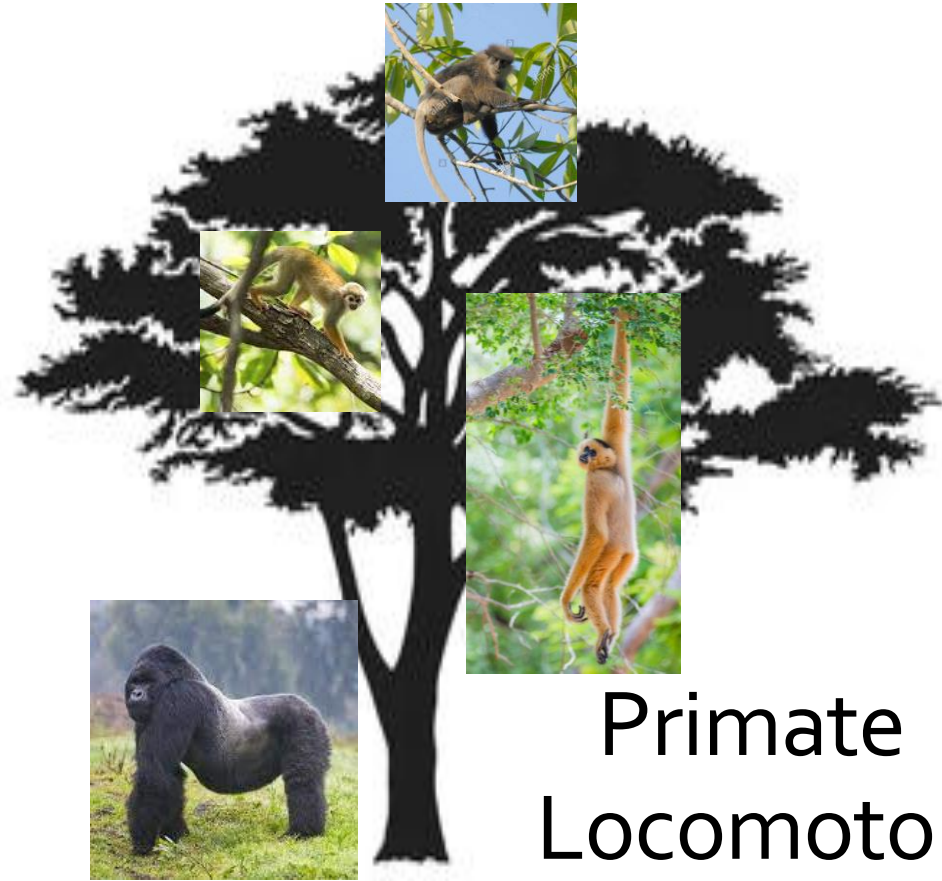
Climbers



Why do we see this locomotor diversity?

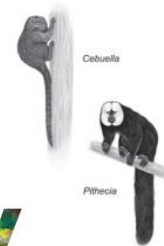
1. **Different environments present different selection pressures**
 - The trait must increase an individual's **fitness** for it to be adaptive, meaning the benefits must outweigh the cost
2. **There is natural morphological variation for evolution to act upon**
 - "Survival of the fittest"

Form ↔ Function



Primate Locomotor Diversity

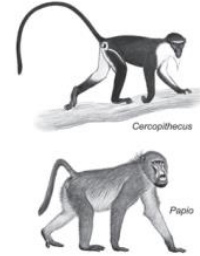
Leapers



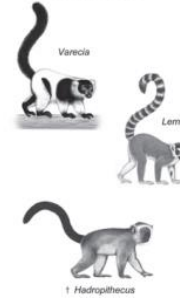
Leapers

Quadrupeds

Eurasia & Africa



Madagascar

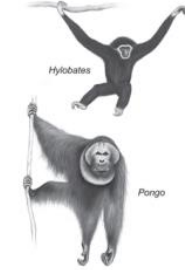


New World



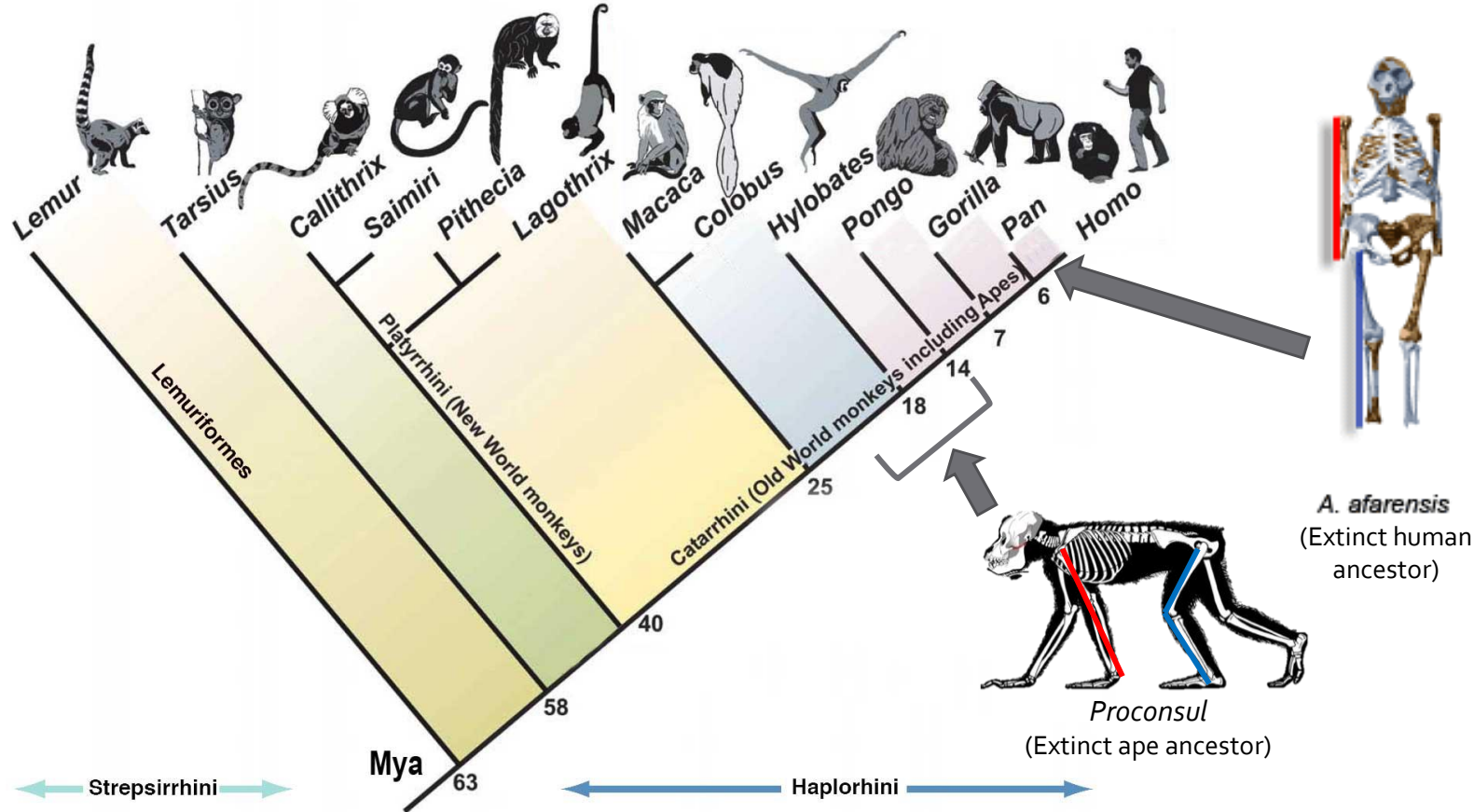
Quadrupeds

Hangers



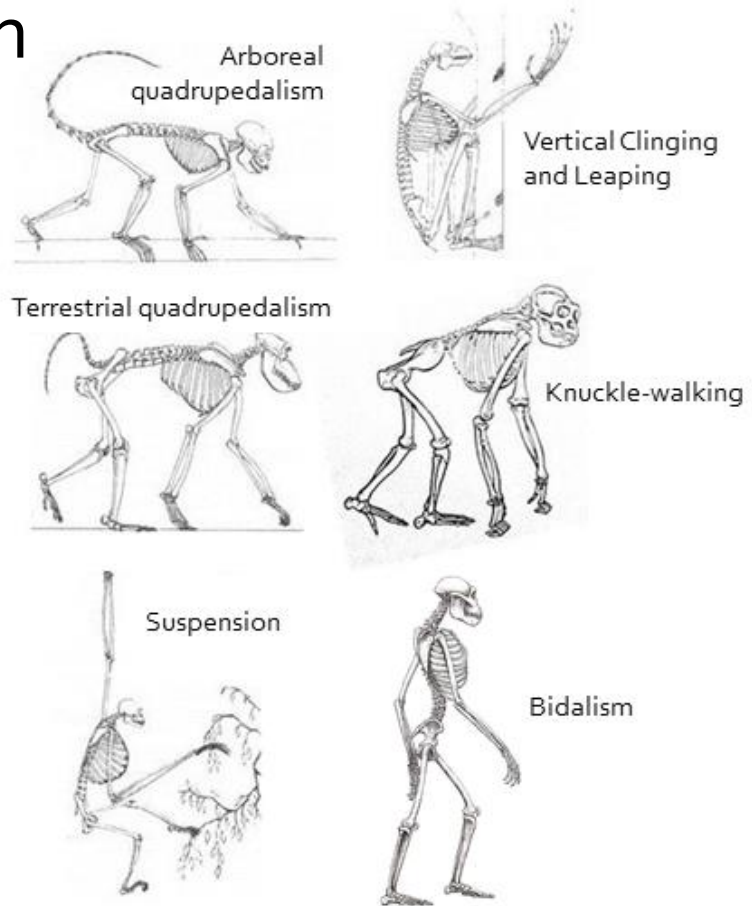
Hangers

Primate evolution – inferring the past from the present



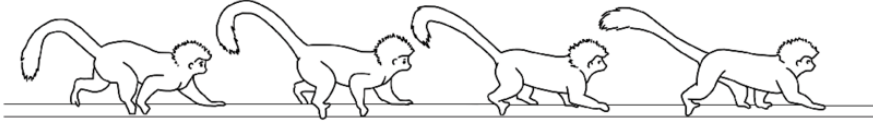
Modes of primate locomotion

- **Arboreal quadrupedalism** -- walking on four legs on top of tree branches
- **Terrestrial quadrupedalism** -- walking on four legs on the ground
 - Includes knuckle-walking
- **Vertical Clinging and Leaping (VCL)** - leaping and jumping in trees
- **Suspension** – hanging or swinging in trees
- **Bipedalism** -- walking on two legs

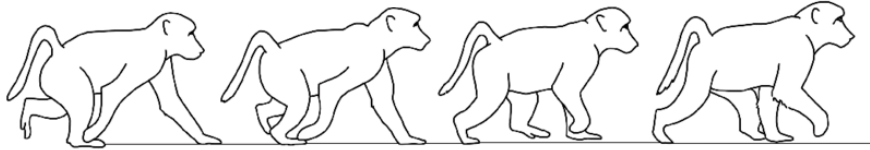


Modes of primate locomotion

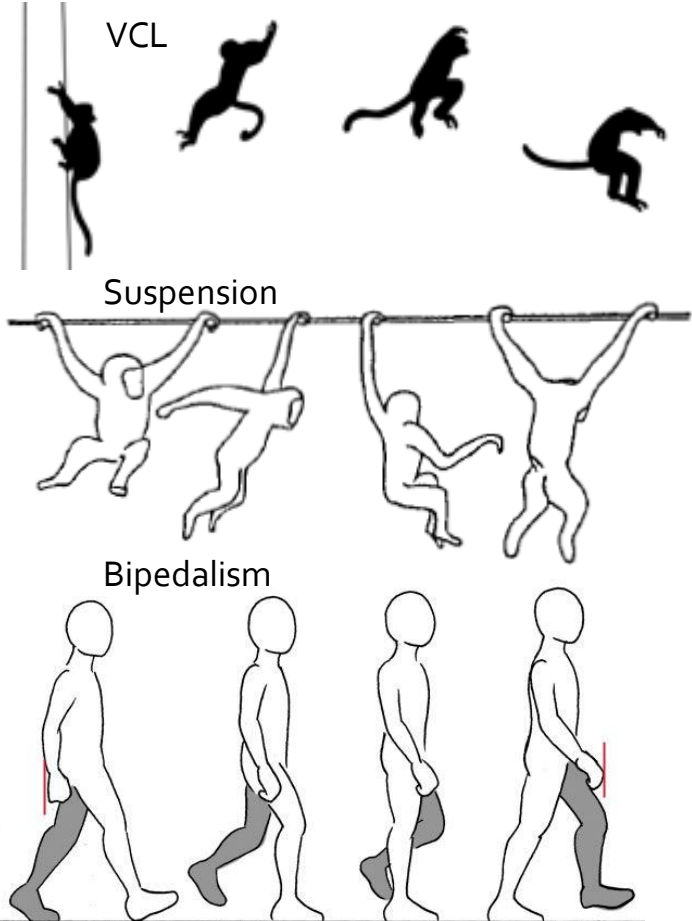
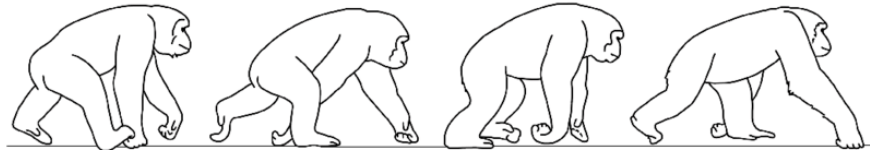
Arboreal quadrupedalism



Terrestrial quadrupedalism

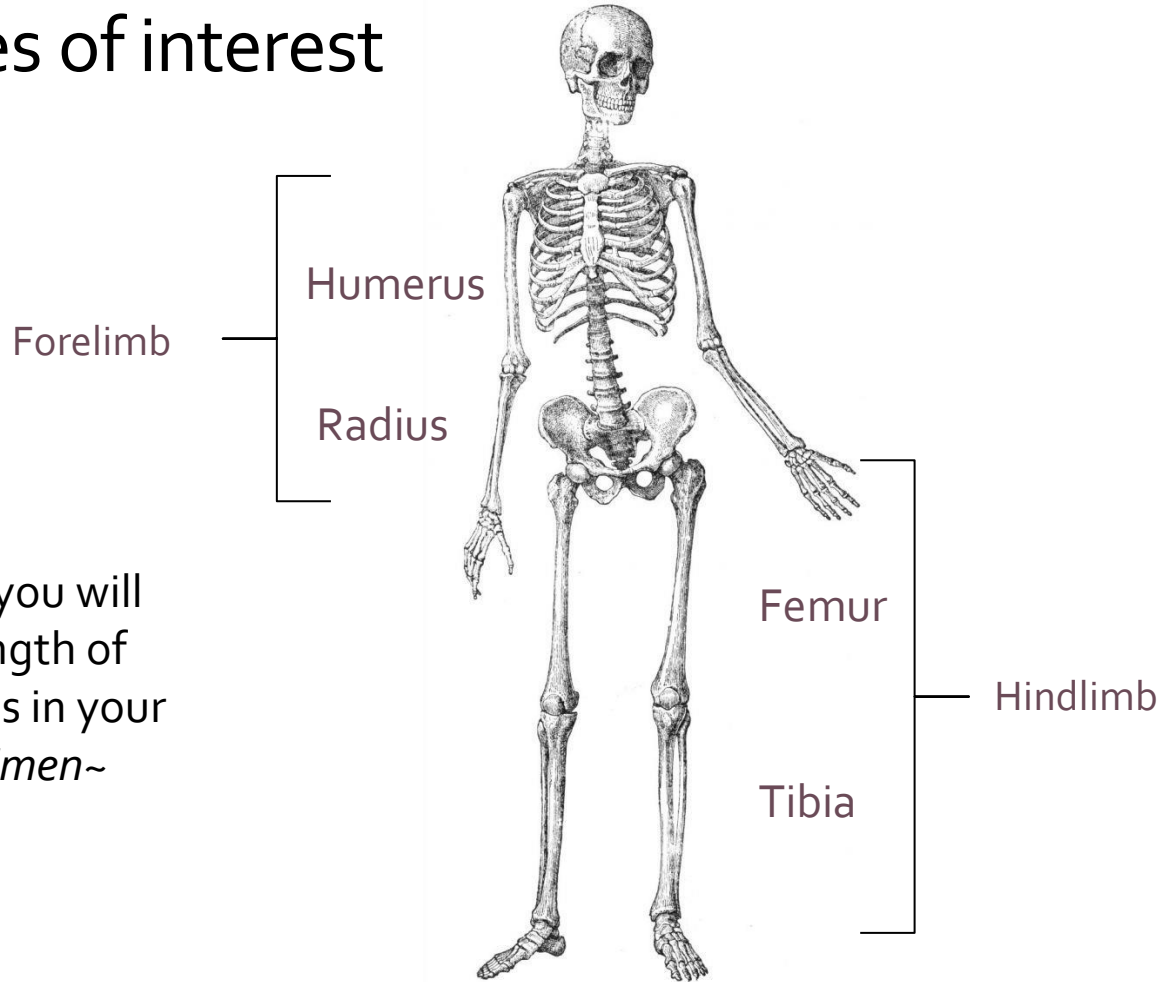


Knuckle-walking (Terrestrial quadrupedalism)



Review: Bones of interest

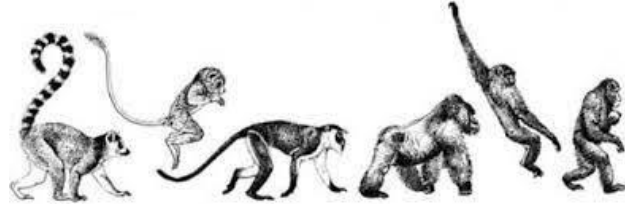
For the activity, you will
measure the length of
these 4 long bones in your
~Mystery Specimen~





Activity portion

Making observations and forming hypotheses



Thinking about the relationship between **structure and function**, take note of the mode of locomotion, the relative length of the hindlimb and forelimb, and anything else notable.

What patterns do you observe?



Part A



Primate
video 1



Primate video 2

Smithsonian
CHANNEL



Primate
video 3



Primate
video 4

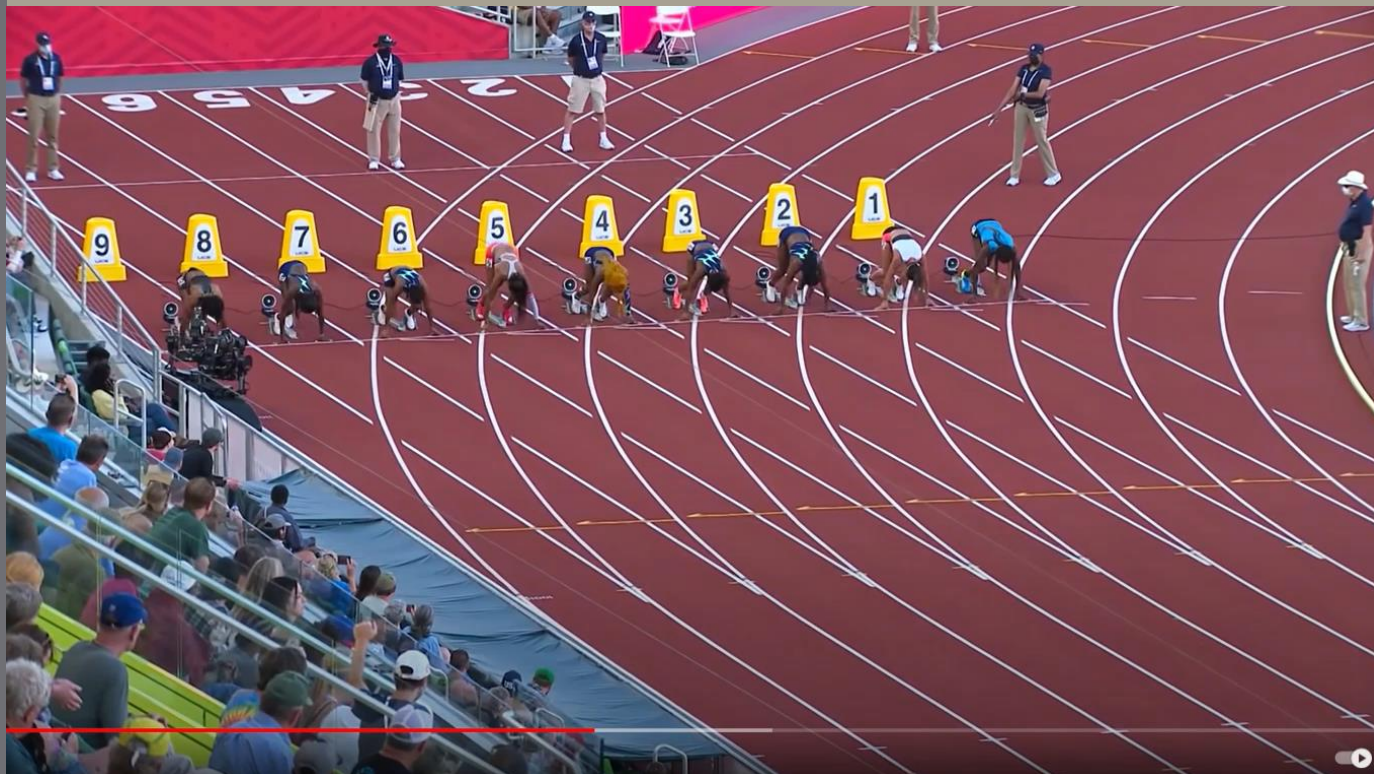
BBC



Primate
video 5

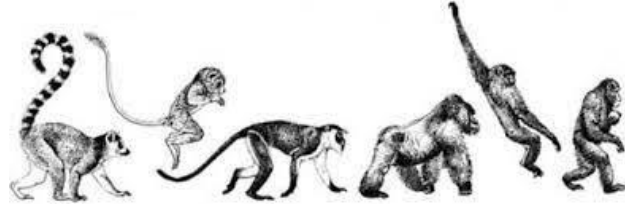


Primate
video 6



Primate
video 7

Making observations and forming hypotheses



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What patterns do you observe?



Part A



How can we compare hindlimb and forelimb lengths across species?

Body size varies across individuals and especially across taxa, so raw limb bone length measurements are not useful for direct comparisons...

Solution?



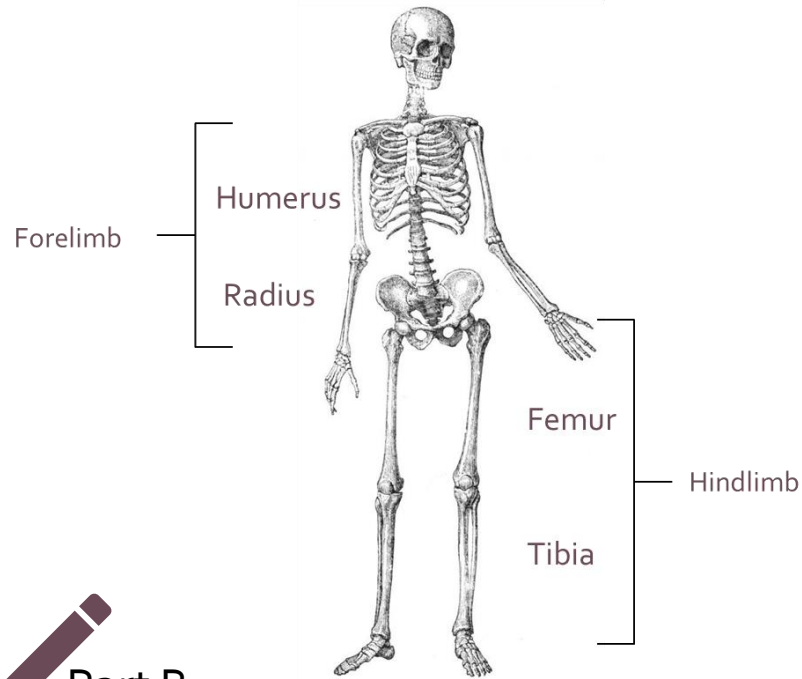
Relative limb proportions	IM Index ?	Mode(s) of locomotion
Longer forelimbs than hindlimbs	High IMI (>100)	
Roughly equal forelimbs and hindlimbs	Intermediate IMI (~100)	
Longer hindlimbs than forelimbs	Low IMI (<100)	



Part B

What index should we expect for each set of relative limb proportions?

$$\text{Intermembral index} = \frac{(\text{humerus} + \text{radius}) * 100}{(\text{femur} + \text{tibia})}$$



Workbook practice:



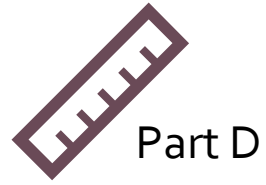
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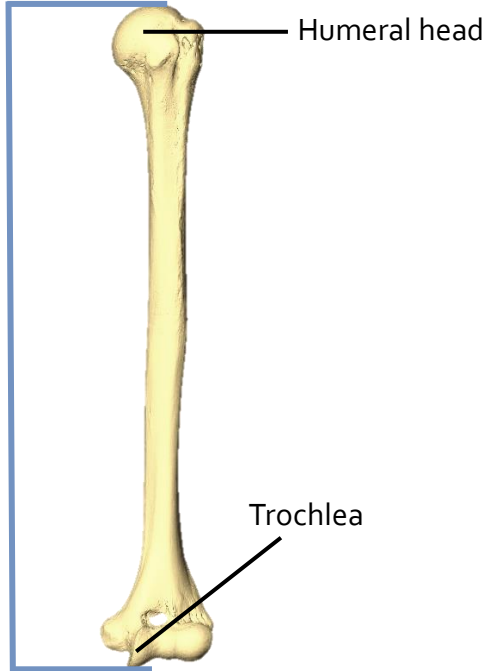
Part C

What modes of locomotion do you *hypothesize* are associated with each intermembral index (IMI)?

Measuring limb bone lengths: Forelimb



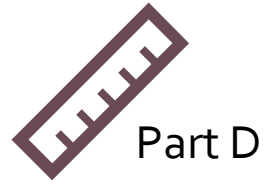
Humerus = The humeral head to the trochlea



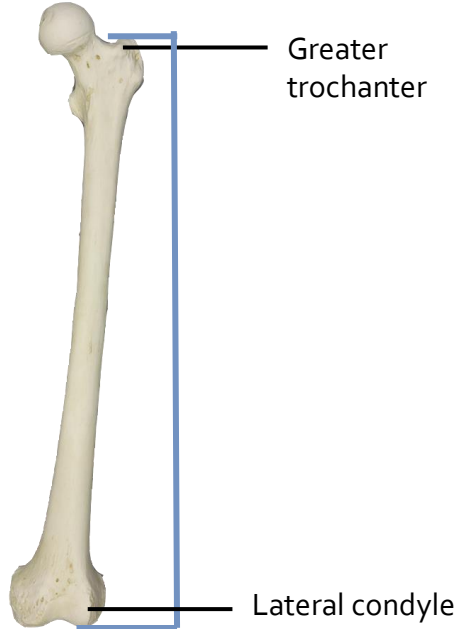
Radius = The radial head to the tip of the styloid process



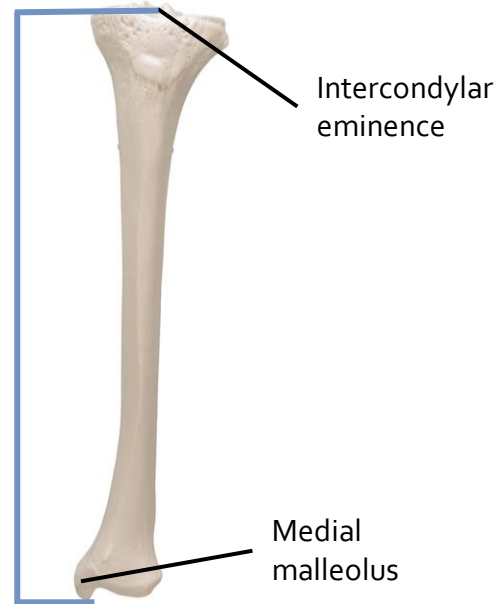
Measuring limb bone lengths: Hindlimb



Femur = The greater trochanter to the lateral condyle



Tibia = The intercondylar eminence to the tip of the medial malleolus



Thank you!

Questions before the Morphosource tutorial?